

Medical Cost Prediction

Executed project report with saved notebook outputs, tables, and charts

Project Description

Medical Cost Prediction

Project Summary

This project is a clean, student-friendly implementation of a medical cost regression. The main goal is to predict medical insurance charges from profile and health fields. The notebook keeps the workflow simple: load the dataset, understand the columns, clean the data, train a baseline model, and check how well the model performs.

No external repository links or profile links are included in this description. Everything needed to understand and run the project is kept inside this folder.

Problem Statement

A raw dataset is useful only when it is converted into a clear decision-making workflow. In this project, the data is processed step by step so that important patterns can be found and a machine learning model can be trained. The expected output is based on charges.

Files Included

File	Purpose	Quick Note
insurance.csv	Main data source for this folder	1338 sampled rows x 7 columns
Notebook file	Complete Python workflow	Includes loading, cleaning, training, and evaluation
PDF report	Human-readable project summary	Matches the updated project structure

Important Columns

Column	Role
age	Input feature used in the modelling workflow
sex	Input feature used in the modelling workflow
bmi	Input feature used in the modelling workflow
children	Input feature used in the modelling workflow
smoker	Input feature used in the modelling workflow
region	Input feature used in the modelling workflow
charges	Target / output column used in the modelling workflow

Workflow Followed

1. Load the data from the local dataset file present in the same project folder.
2. Inspect the structure using shape, column names, missing values, and basic statistics.
3. Clean the dataset by removing empty columns, handling missing values, and keeping only useful features.

4. Prepare the model input by separating features and the target column.
5. Train a baseline model using a simple scikit-learn pipeline with preprocessing.
6. Evaluate the result using practical metrics such as accuracy, classification report, MAE, RMSE, or R2 score depending on the target type.
7. Write observations in a short and direct way so the project can be explained easily.

How to Run

1. Open the notebook in Jupyter Notebook, JupyterLab, Google Colab, or VS Code.
2. Keep the dataset files in the same folder as the notebook.
3. Run the cells from top to bottom.
4. Read the printed metrics and notes at the end before making conclusions.

Final Note

This version keeps the logic understandable and avoids unnecessary complexity. The aim is not to make the notebook look overloaded, but to make it readable, explainable, and useful for a student-level data science portfolio.

Executed Notebook Outputs

The outputs below were generated from the saved notebook after running the code cells in this project folder.

Cell 2 - 2. Load the dataset

```
Dataset shape: (1338, 7)

age sex bmi children smoker region charges
0 20 female 28.5289 1 yes southwest 16884.92400
1 17 male 33.3015 0 no southeast 1725.55230
2 28 male 32.6082 4 no southeast 4449.46200
3 34 male 22.3109 1 no northwest 21984.47061
4 31 male 28.9560 0 no northwest 3866.85520
```

Cell 3 - 3. Understand the data

```
Rows: 1338
Columns: 7

column dtype missing_values unique_values
0 age int64 0 49
1 sex object 0 2
2 bmi float64 0 1334
3 children int64 0 7
4 smoker object 0 2
5 region object 0 4
6 charges float64 0 1337
```

Cell 4 - 4. Clean the data

```
Shape after simple cleaning: (1338, 7)

age sex bmi children smoker region charges
0 20 female 28.5289 1 yes southwest 16884.92400
1 17 male 33.3015 0 no southeast 1725.55230
2 28 male 32.6082 4 no southeast 4449.46200
3 34 male 22.3109 1 no northwest 21984.47061
4 31 male 28.9560 0 no northwest 3866.85520
```

Cell 5 - 5. Select the target column

```
Selected target column: charges

0 16884.92400
1 1725.55230
2 4449.46200
3 21984.47061
4 3866.85520
Name: charges, dtype: float64
```

Cell 6 - 6. Quick EDA

```
Numeric columns: 4
Categorical columns: 3
No missing values found in the previewed columns.

count mean std min 25% \
age 1338.0 39.248132 14.070599 17.0000 27.000000
bmi 1338.0 30.668515 6.114737 15.9856 26.222925
children 1338.0 1.256353 1.271361 0.0000 0.000000
charges 1338.0 13270.422265 12110.011237 1121.8739 4740.287150

50% 75% max
age 39.00000 51.000000 65.00000
bmi 30.33435 34.685750 52.50540
children 1.00000 2.000000 6.00000
charges 9382.03300 16639.912515 63770.42801
```

Cell 7 - 7. Prepare features and target

```
Dropped ID-like columns: ['bmi']
Using 1200 sampled rows for model training speed.
Problem type: regression
Feature shape: (1200, 5)
Target unique values: 1199
```

Cell 8 - 8. Train a baseline model

```
Training completed.
```

Cell 9 - 9. Evaluate the model

```
MAE : 4520.9806
RMSE: 7340.0130
R2 : 0.6227
```

